System
searchPatient(searchCriteria) selectPatientNDService(patientID, medicalService) selectAppointment(professionalID, date, time) verifyReferral(referralID) confirmAppointment()

Contract CO1: searchPatient
Operation: searchPatient(searchCriteria: String)
Cross References: Use Case: Book Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.

Postconditions:
This operation doesn't change the state of the system.

Contract CO2: selectPatientNDService
Operation: selectPatientNDService(patientID: Integer, medicalService: String)
Cross References: Use Case: Book Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.
- *patientID* already exists in the database.
- *medicalService* already exists in the database.

Postconditions:
This operation doesn't change the state of the system.

Contract CO3: selectAppointment
Operation: selectAppointment(professionalID: Integer, date: Date, time: Time)
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *professionalID* already exists in the database.
- A time slot *ts* for that professional at the given time and date already exists in the database.
- *ts.availability* is set to available.

Postconditions:

- An appointment *a* was created. (instance creation)
- *a.status* was set to requested. (attribute modification)
- *a* was associated with the patient's *patientID*. (association formed)
- *a* was associated with the medical service's *medicalService*. (association formed)
- *a* was associated with the professional's *professionalID*. (association formed)
- *a* was associated with *ts*. (association formed)
- *ts.availability* was set to unavailable. (attribute modification)

Contract CO4: verifyReferral
Operation: verifyReferral(referralID: Integer)
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *referralID* already exists in the database.
- The medical service needs a referral.

Postconditions:

- *a* was associated with the referral's *referralID*. (association formed)

Contract CO5: confirmAppointment
Operation: confirmAppointment()
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *a.status* is set to requested.
- *a* is associated with a patient that exists in the database.
- *a* is associated with a medical service that exists in the database.
- *a* is associated with a professional that exists in the database.
- *a* is associated with a time slot *t* that exists in the database.
- *a* is associated with a referral that exists in the database. (onl
- *t.availability* is set to available.

Postconditions:

- *a.status* was set to confirmed. (attribute modification)

System
openPatientRecord(patientID) recordClinicalInformation(diagnosis, observation, treatment, followUpRequirements) issuePrescription(medications) requestFollowUp(followUpRequirements) endConsultation()

Contract CO6: openPatientRecord
Operation: openPatientRecord(patientID: Integer)
Cross References: Use Case: Patient Consultation
Preconditions:

- The user is authenticated, and their role allows access to clinical information.
- *patientID* already exists in the database.

Postconditions:

- A consultation *c* was created. (instance creation)
- *c.startTime* was set to the current time and date. (attribute modification)
- *c.status* was set to in progress. (attribute modification)

- *c* was associated with the patient's *patientID*. (association formed)
- *c* was associated with the professional's *professionalID*. (association formed)

Contract CO7: recordClinicalInformation
Operation: recordClinicalInformation(diagnosis: String, observation: String, treatment: String, followUpRequirements: String)
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- *c.diagnosis* was set to diagnosis via user input. (attribute modification)
- *c.observation* was set to observation via user input. (attribute modification)
- *c.treatment* was set to treatment via user input. (attribute modification)
- *c.followUpRequirements* was set to follow up requirements via user input. (attribute modification)

Contract CO8: issuePrescription
Operation: issuePrescription(medications: List[String])
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- A prescription *p* was created. (instance creation)
- *p* was associated with *c*. (association formed)
- *p.medications* was set to medications (attribute modifications)
- *p.issueDate* was set to the current time and date. (attribute modifications)

Contract CO9: requestFollowUp
Operation: requestFollowUp(followUpRequirements: String)
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- A follow up request *f* was created. (instance creation)
- *f* was associated with *c*. (association formed)
- *f.requirements* was set followUpRequirements. (attribute modifications)
- *f.status* was set to pending. (attribute modifications)

Contract C10: endConsultation
Operation: endConsultation()
Cross References: Use Case: Patient Consultation

Preconditions:

- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:

- *c.endTime* was set to the current time and date. (attribute modifications)
- *c.status* was set to completed. (attribute modifications)

System
searchPatient(searchCriteria) selectPatientAndService(patientID, medicalService) generateInvoice() recordPayment(invoiceID, amount, paymentMethod)

Contract C11: searchPatient
Operation: searchPatient(searchCriteria: String)
Cross References: Use Case: Payment Process
Preconditions:

- The user is authenticated, and their role allows appointment editing.

Postconditions:
This operation doesn't change the state of the system.

Contract C12: selectPatientAndService
Operation: selectPatientAndService(patientID: Integer, medicalService: String)
Cross References: Use Case: Payment Process
Preconditions:

- The user is authenticated, and their role allows appointment editing.
- *patientID* already exists in the database.
- *medicalService* already exists in the database.

Postconditions:
This operation doesn't change the state of the system.

Contract C13: generateInvoice
Operation: generateInvoice()
Cross References: Use Case: Payment Process
Preconditions:

- The patient is not insured so they must pay.

Postconditions:

- An invoice *i* was created. (instance creation)
- *i.total* was set to the service's fee. (attribute modifications)
- *i.status* was set to unpaid. (attribute modifications)
- *i.issueDate* was set to the current time and date. (attribute modifications)
- *i* was associated with the patient's *patientID*. (association formed)

Contract C14: recordPayment
Operation: recordPayment(invoiceID: Integer, amount: Double, paymentMethod: String)
Cross References: Use Case: Payment Process
Preconditions:

- Invoice *i* already exists in the database.
- *i.status* is set to unpaid.

Postconditions:

- A payment *p* was created. (instance creation)
- *p.amount* was set to amount. (attribute modifications)
- *p.paymentMethod* was set to paymentMethod. (attribute modifications)
- *p.date* was set to the current time and date. (attribute modifications)
- *p* was associated with *i*. (association formed)
- *i.status* was set to paid. (association formed)